Gmail Harith Al-Safi <harith.alsafi@gmail.com>

Fwd: Update

3 messages

Alsafadi, Abdallah <abdallah.alsafadi.23@ucl.ac.uk> 5 June 2024 at 22:29
To: "harith.alsafi@gmail.com" <harith.alsafi@gmail.com>

Sent from Outlook for iOS

From: Alexander Giles <alexanderkartungiles@gmail.com>
Sent: Wednesday, June 5, 2024 8:26 PM
To: Alsafadi, Abdallah <abdallah.alsafadi.23@ucl.ac.uk>
Subject: Re: Update

⚠ Caution: External sender

Dear Abdallah,

Could you please send through the study you are doing with Dr Grubb.

Attached!

And is it possible to send me the link for where you got the data on Bloomberg.

We downloaded from the Bloomberg laboratory in computer science, you get an account and use the Python API to request the data, it took many months to collect painstakingly. Usually data collection is quite difficult.

Lastly, what is the difference between the zip file for the holders and the excel sheet for the Sustainable Debt.

The zip file contains a list of holders for each bond. The sustainable debt excel sheet gives information on the bonds themselves, but omits the holders.

Let me know if you need any more info!

See what you can do about getting some intrinsic and extrinsic information for each bond holder (e.g. Black Rock). We can meet to discuss this whenever. You effectively need a list of properties for each of the unique parties that can be seen in that bond holders file. Try that instead of the underwriter syndicates for now.

Here is a first exercise, give it a go and let me know? Can meet whenever if stuck! The python part in particular, we can go through it. But try to do what you can on this random example data first, then we move onto the actual data.

Exercise: Link Prediction Using Gradient Boosting on Network Data

Objective: To understand how to use gradient boosting for link prediction in a network.

Steps:

1. **Create a Network:**
   - Construct a simple undirected network with 6 nodes (A, B, C, D, E, F) and some edges. Represent this network with an adjacency matrix.
2. **Generate Features for Link Prediction:**
   - For each possible pair of nodes, generate features that can be used for link prediction. Possible features include:
     - Common neighbors
     - Jaccard coefficient
     - Adamic/Adar index
     - Preferential attachment
3. **Prepare Training and Testing Data:**
   - Split the data into training and testing sets.
   - For training, include both positive examples (existing links) and negative examples (non-existing links).
   - Ensure the testing set also includes both positive and negative examples.
4. **Apply Gradient Boosting:**
   - Use a gradient boosting algorithm to train a model on the training data.
   - Evaluate the model on the testing data.
5. **Interpret the Results:**
   - Analyze the performance of the model using appropriate metrics like accuracy, precision, recall, and F1-score.
   - Interpret the results and discuss the importance of the features used.

Detailed Instructions:

Network Construction:

Create an undirected network with 6 nodes. Use the following adjacency matrix:

mathematica Copy code

```
  A B C D E F
A 0 1 0 1 0 0
B 1 0 1 0 1 0
C 0 1 0 1 0 1
D 1 0 1 0 0 0
E 0 1 0 0 0 1
F 0 0 1 0 1 0
```

1. **Generate Features:**
   - For each pair of nodes, calculate the following features:
     - **Common Neighbors:** Number of common neighbors between two nodes.
     - **Jaccard Coefficient:** Ratio of the number of common neighbors to the number of total unique neighbors.
     - **Adamic/Adar Index:** Sum of the inverse logarithms of the degrees of common neighbors.
     - **Preferential Attachment:** Product of the degrees of the two nodes.
2. **Prepare Data:**
   - Create a dataset where each row represents a pair of nodes and includes the calculated features and a label (1 for existing link, 0 for non-existing link).
   - Split this dataset into training (70%) and testing (30%) sets.
3. **Gradient Boosting Implementation:**
   - Use the following Python code to implement gradient boosting for link prediction (attached to this email).

```
import numpy as np
import pandas as pd
from sklearn.model_selection import train_test_split
from sklearn.ensemble import GradientBoostingClassifier
from sklearn.metrics import accuracy_score, precision_score, recall_score, f1_score

# Define the adjacency matrix
adjacency_matrix = np.array([
    [0, 1, 0, 1, 0, 0],
    [1, 0, 1, 0, 1, 0],
    [0, 1, 0, 1, 0, 1],
```

```
    [1, 0, 1, 0, 0, 0],
    [0, 1, 0, 0, 0, 1],
    [0, 0, 1, 0, 1, 0]
])

nodes = ['A', 'B', 'C', 'D', 'E', 'F']
num_nodes = len(nodes)

# Generate features
def common_neighbors(a, b):
    return np.sum(adjacency_matrix[a] * adjacency_matrix[b])

def jaccard_coefficient(a, b):
    intersection = np.sum(adjacency_matrix[a] * adjacency_matrix[b])
    union = np.sum(adjacency_matrix[a] + adjacency_matrix[b]) - intersection
    return intersection / union if union != 0 else 0

def adamic_adar_index(a, b):
    common = adjacency_matrix[a] * adjacency_matrix[b]
    return np.sum(common / np.log(np.sum(adjacency_matrix, axis=1) + 1))

def preferential_attachment(a, b):
    return np.sum(adjacency_matrix[a]) * np.sum(adjacency_matrix[b])

data = []
labels = []

for i in range(num_nodes):
    for j in range(i + 1, num_nodes):
        data.append([
            common_neighbors(i, j),
            jaccard_coefficient(i, j),
            adamic_adar_index(i, j),
            preferential_attachment(i, j)
        ])
        labels.append(adjacency_matrix[i, j])

df = pd.DataFrame(data, columns=['Common Neighbors', 'Jaccard Coefficient', 'Adamic/Adar Index', 'Preferential Attachment'])
df['Label'] = labels

# Split into training and testing sets
X_train, X_test, y_train, y_test = train_test_split(df.drop('Label', axis=1), df['Label'], test_size=0.3, random_state=0)

# Train Gradient Boosting Classifier
gbc = GradientBoostingClassifier(random_state=0)
gbc.fit(X_train, y_train)

# Predict and evaluate
y_pred = gbc.predict(X_test)

accuracy = accuracy_score(y_test, y_pred)
precision = precision_score(y_test, y_pred)
recall = recall_score(y_test, y_pred)
f1 = f1_score(y_test, y_pred)

print(f"Accuracy: {accuracy}")
print(f"Precision: {precision}")
print(f"Recall: {recall}")
print(f"F1-score: {f1}")
```

5. **Interpret the Results:**
    - Report the accuracy, precision, recall, and F1-score of the model.
    - Discuss which features were most important in predicting links.
    - Reflect on the performance and possible improvements.

Submission:

- Submit the generated features data.
- Submit the trained model and evaluation results.
- Provide a brief interpretation of the results and feature importance.

Example solution:

	Common Neighbors	Jaccard Coefficient	Adamic/Adar Index
0	1	0.25	0.910239
1	1	0.20	0.910239
2	2	0.40	1.820479
3	0	0.00	0.000000
4	0	0.00	0.000000
5	1	0.20	0.910239
6	1	0.25	0.910239
7	1	0.25	0.910239
8	1	0.25	0.910239
9	0	0.00	0.000000
10	1	0.20	0.910239
11	1	0.25	0.910239
14	1	0.20	0.910239
15	1	0.25	0.910239

Model Evaluation Results:

```python
Accuracy: 0.8333
Precision: 0.8571
Recall: 0.7500
F1-score: 0.8000
```

Best Wishes,

Alex

On 5/31/24 19:40, Alsafadi, Abdallah wrote:

> Hey Dr Giles,
> Hope all is well,
> Could you please send through the study you are doing with Dr Grubb. And is it possible to send me the link for where you got the data on Bloomberg. Lastly, what is the difference between the zip file for the holders and the excel sheet for the Sustainable Debt.
> I hope you have a wonderful weekend!
>
> Best,
> Abdallah
> Sent from Outlook for iOS

2 attachments

main.pdf
1464K

code_for_gradient_boosting.txt
3K

Alsafadi, Abdallah <abdallah.alsafadi.23@ucl.ac.uk>
To: "harith.alsafi@gmail.com" <harith.alsafi@gmail.com>

Sent from Outlook for iOS

From: Alexander Giles <alexanderkartungiles@gmail.com>
Sent: Wednesday, June 5, 2024 8:38:03 PM
To: Alsafadi, Abdallah <abdallah.alsafadi.23@ucl.ac.uk>
Subject: Re: Update

```
Empty DataFrame
Columns: []
Index: []
----------------------------------------
data point 1
----------------------------------------
```

	holder_name_	portfolio_name_	parent_holder_Id	holder_id__	portfolio_number_	position_	position_change__	filing_date__	filing_source_	insider_status_	percent_outstanding	percent_of_portfolio	institution_type_	met
BO4675365 Corp	Vanguard Group Inc/The	n/a	0.0	287915.0	0.0	112658.0	-460.0	2023-09-30	ULT-AGG	N-P	1.25	0.0	Investment Advisor	Phila
BO4675365 Corp	Intesa Sanpaolo SpA	Multiple Portfolios	0.0	141800.0	0.0	35086.0	2552.0	2023-07-31	MF-AGG	N-P	0.39	0.0	Bank	
BO4675365 Corp	Etica SGR SpA/Italy	Multiple Portfolios	0.0	170162.0	0.0	35000.0	0.0	2023-04-28	MF-AGG	N-P	0.39	0.0	Investment Advisor	
BO4675365 Corp	Dimensional Fund Advisors LP	Multiple Portfolios	0.0	80528.0	0.0	23774.0	-43.0	2023-09-30	MF-AGG	N-P	0.26	0.0	Investment Advisor	
BO4675365 Corp	Union Investment Luxembourg SA	n/a	0.0	230817.0	0.0	23301.0	1.0	2023-08-31	ULT-AGG	N-P	0.26	0.0	Bank	Uncla
BO4675365 Corp	St James's Place PLC	n/a	0.0	338417.0	0.0	21050.0	15000.0	2023-03-31	ULT-AGG	N-P	0.23	0.0	Holding Company	
BO4675365 Corp	DFA AUSTRALIA LTD	Multiple Portfolios	0.0	338295.0	0.0	19770.0	0.0	2023-09-30	MF-AGG	N-P	0.22	0.0	Unclassified	
BO4675365 Corp	Capital Group Cos Inc/The	n/a	0.0	94115.0	0.0	17100.0	0.0	2023-10-20	ULT-AGG	N-P	0.19	0.0	Investment Advisor	Los Angeles/P
BO4675365 Corp	BlackRock Inc	n/a	0.0	227843.0	0.0	12525.0	0.0	2023-11-09	ULT-AGG	N-P	0.14	0.0	Investment Advisor	New York City/Southern CT/Nort
BO4675365 Corp	JPMorgan Chase & Co	n/a	0.0	120467.0	0.0	9600.0	0.0	2023-06-30	ULT-AGG	N-P	0.11	0.0	Bank	New York City/Southern CT/Nort
BO4675365 Corp	SILEX FUND SICAV	Multiple Portfolios	0.0	448463.0	0.0	9000.0	0.0	2023-06-30	MF-AGG	N-P	0.10	0.0	Investment Advisor	Uncla
BO4675365 Corp	RAIFFEISEN BANK INTERNATIONAL	n/a	0.0	341160.0	0.0	8400.0	0.0	2023-09-29	ULT-AGG	N-P	0.09	0.0	Bank	
BO4675365 Corp	Credit Agricole Group	n/a	0.0	161118.0	0.0	6275.0	0.0	2023-11-07	ULT-AGG	N-P	0.07	0.0	Bank	
BO4675365 Corp	Government Pension Investment Fund	Multiple Portfolios	0.0	336518.0	0.0	6000.0	10.0	2023-03-31	MF-AGG	N-P	0.07	0.0	Pension Fund	
BO4675365 Corp	BNP Paribas SA	n/a	0.0	288127.0	0.0	4987.0	72.0	2023-10-31	ULT-AGG	N-P	0.06	0.0	Investment Advisor	Uncla
BO4675365 Corp	Pictet Funds SA	n/a	0.0	284576.0	0.0	4944.0	-200.0	2023-07-31	ULT-AGG	N-P	0.05	0.0	Investment Advisor	
BO4675365 Corp	Allianz SE	n/a	0.0	228194.0	0.0	4800.0	1200.0	2023-09-30	ULT-AGG	N-P	0.05	0.0	Investment Advisor	
BO4675365 Corp	Pramerica Management Co SA	Multiple Portfolios	0.0	133209.0	0.0	4400.0	0.0	2023-07-31	MF-AGG	N-P	0.05	0.0	Investment Advisor	Lux
BO4675365 Corp	KBC Group NV	n/a	0.0	148493.0	0.0	4125.0	-150.0	2023-06-30	ULT-AGG	N-P	0.05	0.0	Bank	B
BO4675365 Corp	Wellington Management Group LLP	Multiple Portfolios	0.0	81060.0	0.0	4110.0	0.0	2023-09-30	MF-AGG	N-P	0.05	0.0	Investment Advisor	
BO4675365 Corp	Morgan Stanley	n/a	0.0	228201.0	0.0	4000.0	0.0	2023-09-30	ULT-AGG	N-P	0.04	0.0	Brokerage	New York City/Southern CT/Nort
BO4675365 Corp	HSBC Holdings PLC	n/a	0.0	330613.0	0.0	3840.0	0.0	2023-09-29	ULT-AGG	N-P	0.04	0.0	Investment Advisor	
BO4675365 Corp	UBS AG	n/a	0.0	229311.0	0.0	3000.0	0.0	2023-07-31	ULT-AGG	N-P	0.03	0.0	Investment Advisor	
BO4675365 Corp	Desjardins Global Asset Management	Multiple Portfolios	0.0	50608.0	0.0	2928.0	0.0	2023-09-29	MF-AGG	N-P	0.03	0.0	Investment Advisor	M
BO4675365 Corp	Legg Mason Asset Management Singap	Multiple Portfolios	0.0	300082.0	0.0	2690.0	1600.0	2023-07-31	MF-AGG	N-P	0.03	0.0	Investment Advisor	Uncla

Is this ok?

Also you might need some help loading this text based data into python, I will send a CSV file for the holders.

Best Wishes,

Alex

On 6/5/24 20:28, Alsafadi, Abdallah wrote:

> Dear Dr Giles,
> Thank you so much! Will get on to this pronto. One last question, could you tell me what are the columns for the holders, as they aren't shown.
>
> Thank you once more!
> Best,
> Abdallah
>
> Sent from Outlook for iOS
>
> **From:** Alexander Giles <alexanderkartungiles@gmail.com>
> **Sent:** Wednesday, June 5, 2024 8:23:35 PM
> [Quoted text hidden]
>
> [Quoted text hidden]

Alsafadi, Abdallah <abdallah.alsafadi.23@ucl.ac.uk>
To: "harith.alsafi@gmail.com" <harith.alsafi@gmail.com>

5 June 2024 at 22:51

Sent from Outlook for iOS

From: Alexander Giles <alexanderkartungiles@gmail.com>
Sent: Wednesday, June 5, 2024 8:39:39 PM
To: Alsafadi, Abdallah <abdallah.alsafadi.23@ucl.ac.uk>
Subject: Re: Update

Attached is CSV of the holders data.

Any help let me know.

[Quoted text hidden]

holdersbonds_all_Feb.csv
18032K